

LAB ACTIVITY: Implementing a Secure OpenVPN Tunnel

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(ekfm2025@kali)-[~/openvpn-ca]
$ sudo systemctl status openvpn-server@server
● openvpn-server@server.service - OpenVPN service for server
   Loaded: loaded (/usr/lib/systemd/system/openvpn-server@.service; disabled; preset: disabled)
   Active: active (running) since Thu 2026-05-14 22:15:29 PST; 25s ago
     Invocation: e5765e4df0ac46408c2238a50dc706a3
       Docs: man:openvpn(8)
             https://openvpn.net/community-resources/reference-manual-for-openvpn-2-7/
             https://community.openvpn.net/openvpn/wiki/HOWTO
    Main PID: 214269 (openvpn)
      Status: "Initialization Sequence Completed"
        Tasks: 1 (limit: 20)
       Memory: 2.2M (peak: 2.3M)
          CPU: 17ms
      CGroup: /system.slice/system-openvpn\x2dservice.slice/openvpn-server@server.service
              └─214269 /usr/sbin/openvpn --status /run/openvpn-server/status-server.log --status-version 2 --suppre>

May 14 22:15:29 kali openvpn[214269]: WARNING: --keepalive option is missing from server config
May 14 22:15:29 kali openvpn[214269]: TUN/TAP device tun0 opened
May 14 22:15:29 kali openvpn[214269]: tun/tap device [tun0] opened
May 14 22:15:29 kali openvpn[214269]: net_iface_mtu_set: mtu 1500 for tun0
May 14 22:15:29 kali openvpn[214269]: net_iface_up: set tun0 up
May 14 22:15:29 kali openvpn[214269]: net_addr_v4_add: 10.8.0.1/24 dev tun0
May 14 22:15:29 kali openvpn[214269]: Could not determine IPv4/IPv6 protocol. Using AF_INET
May 14 22:15:29 kali openvpn[214269]: UDPv4 link local (bound): [AF_INET][undef]:1194
May 14 22:15:29 kali openvpn[214269]: UDPv4 link remote: [AF_UNSPEC]
May 14 22:15:29 kali openvpn[214269]: Initialization Sequence Completed
lines 1-25/25 (END)
```

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(ekfm2025@kali)-[~/openvpn-ca]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
10.8.0.0/24 dev tun0 proto kernel scope link src 10.8.0.1
192.168.1.0/24 dev tun0 scope link

(ekfm2025@kali)-[~/openvpn-ca]
$
```

1. UDP vs TCP: Why is UDP preferred for VPNs? (Hint: Research "TCP Meltdown" or "TCP over TCP").

- UDP is preferred for VPNs because it avoids the performance disaster known as TCP Meltdown or TCP over TCP. When you run a TCP-based VPN tunnel, both the internal data and the tunnel itself try to manage error correction and retransmissions. If a single packet is lost, both layers start a "retry" loop at the same time, which creates an exponential backup that eventually causes the entire connection to collapse. Using UDP allows the VPN to simply transport the data without adding its own layer of overhead, letting the internal applications handle their own error correction much more efficiently

2. PKI Security: If an employee leaves the company, we have their certificate, but they still have a copy on their laptop. How does a Certificate Revocation List (CRL) solve this?

- If an employee leaves the company, a Certificate Revocation List (CRL) acts as a "blacklist" or a "security checkpoint" that the VPN server consults during every login attempt. Even though the employee still has a physical copy of the certificate on their laptop, the administrator adds that specific certificate's serial number to the CRL. When

the ex-employee tries to connect, the server sees their ID is on the "revoked" list and immediately rejects the connection, rendering their local copy of the certificate useless for gaining network access.

3. Visibility: When the client pings 192.168.1.100, does the research server see the client's real IP or the VPN IP (10.8.0.6)? Why?.

- When the client pings 192.168.1.100, the research server will see the VPN IP (10.8.0.6) rather than the client's real IP address. This happens because the VPN server acts as a gateway that "encapsulates" the client's traffic, making it appear as if the request originated from within the trusted virtual tunnel. This provides an essential layer of security by masking the client's true location and physical network details from the internal servers.